Evolveit Tech Network Design (Project 2)

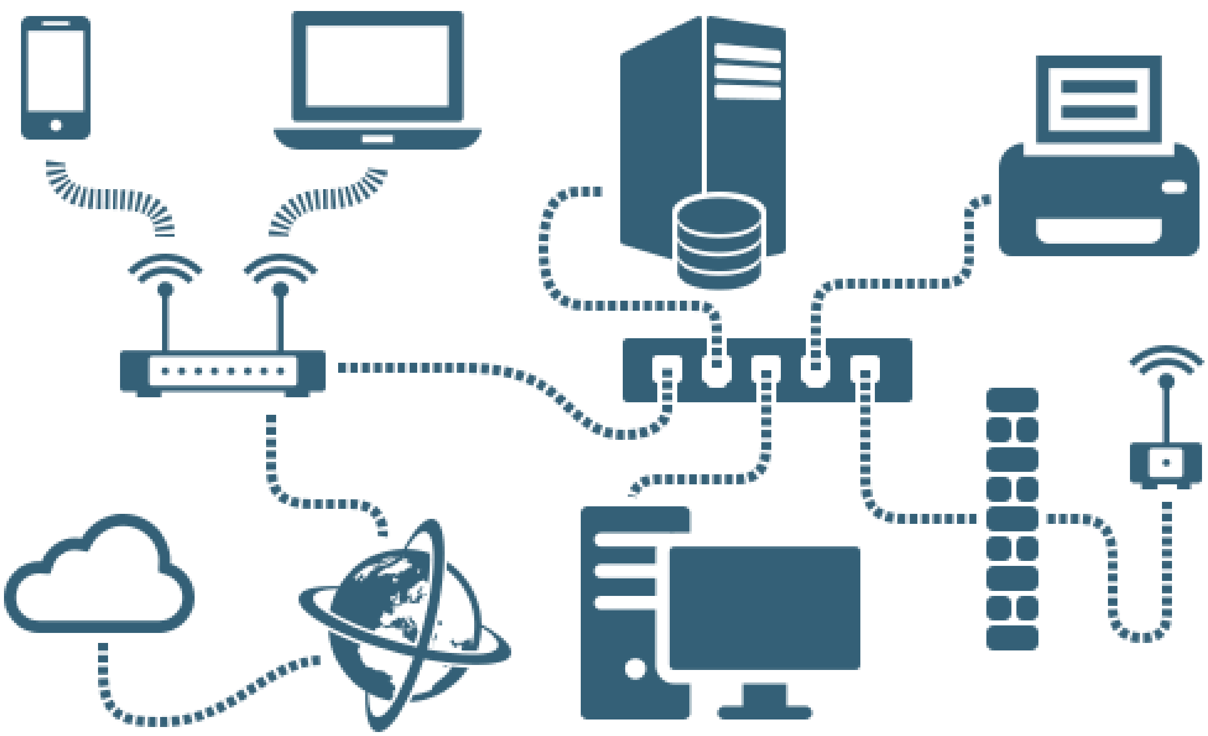

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Table of Contents

Company Network Design

Plan Date: 04/09/2025 – 04/29/2025

Company name: EvolveIT Tech

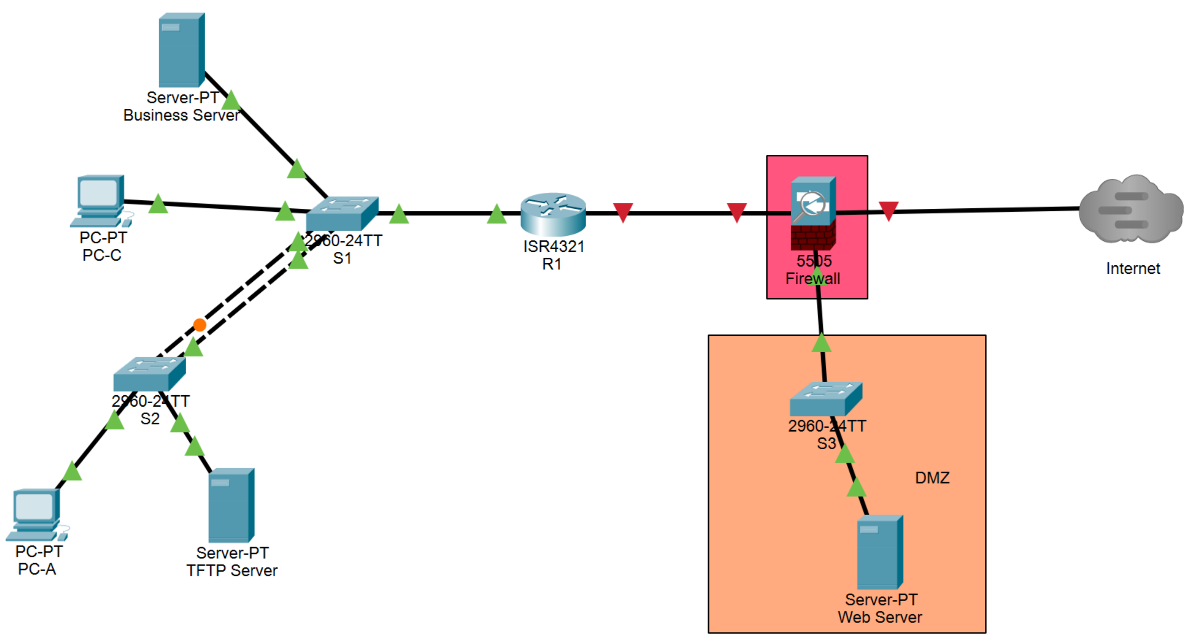
Network Engineer: Hashan Kodippilige

Description of Company: Providing Computer spare parts and support services.

List of Departments:

Management Department	12 Hosts
Operations Department	35 Hosts
Marketing Department	200 Hosts

Topology Diagram:



Subnet List

Network Name	Number of Hosts	Network Mask	Network Address	1st Address	Last Address	Broadcast Address
Marketing	200	255.255.255.0	192.168.1.0	192.168.1.1	192.168.1.254	192.168.1.255
Operations	35	255.255.255.192	192.168.2.0	192.168.2.1	192.168.2.62	192.168.2.63
Management	12	255.255.255.240	192.168.3.0	192.168.3.1	192.168.3.14	192.168.3.15
DMZ	5	255.255.255.192	192.168.4.0	192.168.4.1	192.168.4.62	192.168.4.63
R1 to ASA	2	255.255.255.252	192.168.5.252	192.168.5.253	192.168.5.254	192.168.5.255

Installation/Configuration

1. Install Wirshark

a. Download 64-bit (For Windows 64-bit version) version file from www.wireshark.org to install in the PCs.
b. Double clicks the file and start the installation process.
c. Follow up the steps and if you already have a copy of Wireshark Click **Yes** to uninstall and previous version.
d. In Wireshark Setup Wizard Click **Next**
e. Click **I Agree** when the License Agreement Window displays.
f. On the Choose component window, keep the default settings and Click **Next**
g. Choose your desired shortcut options and click **Next**
h. On the installation location window, keep the original location and click **Next**
i. To capture live network data, WinPcap must be installed on your PC. If WinPcap is already installed on your PC, the Install check box will be unchecked. If you have an older version recommended that you allow the newer version to be installed by clicking the Install WinPcap x.x.x (version number) check box.
j. Do not install USBPcap for normal traffic capture. Do Not Select the checkbox to install USBPcap (It is experimental).
k. Wireshark starts installing its files and displays with the status of the installation. Click **Next** when Wireshark installation is complete and then click **Finish** to complete the process.
l. In a separate window, accept the license agreement in the Npcap Setup Wizard if installing Npcap. Click **I Agree** to continue and then **Next** to finish the Npcap installation and click **Finish** to exit the Npcap installation.

2. Install Putty

a. In the web browser, go to ninite.com
b. Under the Development Tools, select Putty
c. Click **Get your Ninite** button at the bottom of the window.
d. After downloading, open the installer and run it.

3. Install TFTP server

a. Go to www.solarwinds.com/free-tools/free-tftp-server
b. Click **Download free tool**
c. First Name: Hashan   Last Name: Kodippilige
d. Business Email: hashan@hsk.com
e. Company: EvolveIT Tech
f. Zip Code: 58203
g. Business Phone: 7014108095
h. Install
i. Continue without adding
j. Windows-Download now
m. After you download, double click the file and start the installation process.

4. Saving configurations to TFTP server after each Session

4.1 Switches (Do it on each switch) and Router

Switches	Router (R1)
S1# copy flash:vlan.dat tftp Destination IP: 192.168.3.2 File name: s1-vlan.dat Save the configuration S1# copy running-config tftp Destination IP: 192.168.3.2 File name: s1-confg Save the configuration S1# delete vlan.dat S1# erase startup-config	R1# copy running-config tftp Destination IP: 192.168.3.2 File name: r1-confg Save the configuration R1# erase startup-config

5. Restoring configurations from TFTP server

5.1 Switches (Do it for each switch) and Router

Switches	Router (R1)
switch# copy tftp flash:vlan.dat (Answer questions) File name: s1-vlan.dat switch # copy tftp running-config (Answer questions) File name: s1-confg Save the configuration	R1# copy tftp: running-config tftp (Answer questions) File name: r1-confg Save the configuration

Devices and Connections

Device Ports	Connected to Device Ports	Purpose
R1 G0/0/1	S1 F0/5	Trunk
R1 G0/0/0	ASA Firewall G1/2	
S1 F0/1-2	S2 F0/1-2	Trunk
S2: F0/6-17	Management Group	VLAN 20
S1: F0/6-14	Operations Group	VLAN 30
S1: F0/15-24	Marketing Group	VLAN 40
S3:F0/1	ASA Firewall G1/3	Uplink to ASA Firewall
ASA Firewall G1/1	ISP	
S3: F0/1, F0/6	Web Server	VLAN 50

Basic Configurations for Network Devices (See Appendix -1)

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0/0	192.168.5.253	255.255.255.252	N/A
	G0/0/1	N/A	N/A	
	G0/0/1.20	192.168.3.1	255.255.255.240	
	G0/0/1.30	192.168.2.1	255.255.255.192	
	G0/0/1.40	192.168.1.1	255.255.255.0	
	G0/0/1.1000	N/A	N/A	
ASA	G1/1 (OUTSIDE)	172.24.104.4	255.255.255.248	N/A
	G1/2 (INSIDE)	192.168.5.254	255.255.255.252	
	G1/3 (DMZ)	192.168.4.1	255.255.255.192	
S1	VLAN 30	DHCP	255.255.255.192	192.168.2.1
S2	VLAN 20	DHCP	255.255.255.240	192.168.3.1
S3	VLAN 50	192.168.4.2	255.255.255.192	192.168.4.1
TFTP Server	NIC	192.168.3.2	255.255.255.240	192.168.3.1
	NIC & ens33 -Ethernet	192.168.4.3 & 192.168.4.4	255.255.255.192	192.168.4.1
	ens33.0 -Intranet	192.168.4.45	255.255.255.192	192.168.4.1
	enss33.1 -Internet	192.168.4.46	255.255.255.192	192.168.4.1
	ens33.2 -B2B	192.168.4.47	255.255.255.192	192.168.4.1
Business Server	NIC	192.168.2.2	255.255.255.192	192.168.2.1
PC-(3.1-3.11)	NIC	192.168.3.4 – 192.168.3.14	255.255.255.240	192.168.3.1
PC-(2.1-2.34)	NIC	192.168.2.4 – 192.168.2.37	255.255.255.192	192.168.2.1
PC-(1.1-1.200)	NIC	192.168.1.5 – 192.168.1.204	255.255.255.0	192.168.1.1

1. **Router**
    a. Assign a device name to the router.
    b. Disable DNS lookup to prevent the router from attempting to translate incorrectly entered commands as though they were host names.
    c. Assign class as the privileged EXEC encrypted password.
    d. Assign cisco as the console password and enable login.
    e. Assign cisco as the VTY password and enable login.
    f. Encrypt the plaintext passwords.
    g. Create a banner that warns anyone accessing the device that unauthorized access is prohibited.
    h. Save the running configuration to the startup configuration file.

2. **ASA Firewall**
    a. Assign a device name to the firewall.
    b. Assign domain name as hsk-spare.com
    c. Assign cisco as login password.
    d. Create a banner that warns anyone accessing the device that unauthorized access is prohibited.
    e. Enable http server on INSIDE network.
    f. Save the running configuration to the startup configuration file.

3. **Switches**
    a. Assign a device name to the switch.
    b. Disable DNS lookup to prevent the router from attempting to translate incorrectly entered commands as though they were host names.
    c. Assign class as the privileged EXEC encrypted password.
    d. Assign cisco as the console password and enable login.
    e. Assign cisco as the VTY password and enable login.
    f. Encrypt the plain text passwords.
    g. Create a banner that warns anyone accessing the device that unauthorized access is prohibited.
    h. Save the running configuration to the startup configuration file.

4. **PCs**

    a. Assigning of PCs host address will be done through DHCP server and Assign TFTP server, Business server and Web server addresses as above table.

List of Each Feature Configured (See Appendix -2)

VLAN	Name	Interface Assigned
20	Management	S2: F0/6-17
30	Operations	S1: F0/6-14
40	Marketing	S1: F0/15-24
50	DMZ	S3: F0/1 and F0/6
999	ParkingLot	S1: F0/3-4, G0/1-2 S2: F0/3-5, F0/18-24, G0/1-2 S3: F0/2-5, F0/7-24, G0/1-2
1000	Native	N/A

1. **Create VLANs on three switches**
   a. Create VLAN as shown in the table above
   b. Configure the management interface and default gateway on each switch using the IP address information in the Addressing Table.
   c. Assign used ports to the appropriate VLAN (specified in the VLAN table above) and configure them for static access mode.
   d. Assign all unused ports on the switch to the Parking Lot VLAN, configure them for static access mode, and administratively deactivate them.
   e. Verify configuration by issuing **Show Vlan Brief** command.

2. **Configure Trunking**

   2.1 Manually configure trunk interface F0/1-2 on both Switches with EtherChannel

   a. Specify the interfaces using interface range command.
   b. Create the port channel interface with the channel-group identifier mode active command.
   c. To change the layer 2 settings on the port channel interface, enter port channel interface configuration mode using interface port-channel command.
   d. Change the switchport mode on interface to force trunking. Do it on both switches.
   e. Set up Native vlan to 1000 on both switches.
   f. Specify that Vlan 20,30,40 and 1000 are allowed to cross the trunk.
   g. Verify configuration by issuing **Show interface trunk** command.

   2.2 Manually configure S1's trunk interface F0/5

   a. Use the same configuration as F0/1 on switches to configure trunking on interface F0/5 on S1.
   b. Save configuration from Running to Startup.
   c. Verify configuration by issuing **Show interface trunk** command.

3. **<u>PortFast and BPDU Guard Configuration on Switches.</u>**
   a. It can be done either globally or on a per-interface basis. When enable globally, portpast is enabled on all non-trunking ports. Enter the global configuration mode.
   b. Enter **spanning-tree portfast default** command on the global configuration mode.
   c. Enter **spanning-tree bpduguard enable** (Can do both portfast and bpduguard inth same command line using **spanning-tree portfast bpduguard default** command)

4. **<u>Configure Inter-Vlan Routing on Router</u>**
   a. Activate interface G0/0/1 on router.
   b. Configure Sub-interfaces for each VLAN as specified in the IP addressing table. All sun-interfaces use 802.1Q encapsulation. Do not configure IP address for Native VLAN. Include description for each sub-interface.
   c. Verify configuration by issuing **Show ip interface brief** command.

5. **<u>Configure Default Routing for R1</u>**
   a. Configure G0/0/0 to IP address given on the table.
   b. Default route on Router to the ASA interface G1/2

6. **<u>Configure DHCP on Router (R1)</u>**
   Exclude statically assigned addresses for all three vlans separately.
   a. Configure the pool name it with Vlan name and specify the subnet to use when assigning IP addresses.
   b. Configure dns server
   c. Configure the default router and domain name server for the network.
   d. Verify the DHCP configuration.

7. **<u>Configure R1 for PAT</u>**
   a. Define the access list for internal network
   b. Configure the outside interface
   c. Configure inside interfaces
   d. Enable NAT overload (PAT)

8. **<u>Configure all network devices for basic SSH support (R1, and S1-3).</u>**
   a. Create a local user with username SSHadmin and encrypted password @hsk123!
   b. Use hsk-spare.com as domain name.
   c. Generate crypto keys using a 1024-bit modulus.
   d. Configure the first five VTY lines on each device to support SSH connections only and to authenticate to local user database.

9. Configure ASA interfaces

a. Configure the OUTSIDE interface
  - Assign IP address, name the interface, and set the security level to 0 (least trusted)

b. Configure the INSIDE interface
  - Set the IP address and the highest security level (100) for internal LAN access

c. Configure the DMZ interface
  - Assign an IP address, name the interface, and set the security level to 70

10. Configure Default route for the ASA

a. Set astatic default route to the next hop (ISP)

11. Configure address translation using PAT and network objects

a. Create a network object INSIDE-NET
  - Assign attribute to it using the *subnet* and *nat* command.

b. Create a network object DMZ-NET
  - Assign attribute to it using the *subnet* and *nat* command.

12. Enable ICMP Inspection in Global Policy

a. Add the inspection of ICMP traffic to the policy map.
  - Check policy map using *show run policy-map* command

13. Configure Static NAT to the DMZ Server using a network object

a. Create a network object for the Internet webside
  - Define static NAT rules to map it to the outside Public IP

14. Configure an ACL to allow access to the Web server from the Internet

a. Allow OUTSIDE-network to access the Internet website.
  - Allow OUTSIDE access to Internet server (static NAT: 172.24.104.7 to 192.168.4.46)
  - Apply the ACL to the ASA OUTSIDE interface in the *in* direction.

b. Apply the ACL to the ASA to any DMZ host to access any INSIDE host through pinging.
  - Apply the ASA any DMZ host to access any INSIDE host through tcp on HTTP and HTTPS.
  - Apply the ACL to the ASA DMZ interface in the *in* direction.

15. Configure AAA to use local database for authentication and SSH remote access to ASA

a. Create a local user named admin and password cisco12345

b. Configure AAA to use the local ASA database for SSH user authentication.

c. Generate an RSA key as 2048 and save the RSA key to persistent flash memory

d. Configure the ASA to allow SSH connections from any host from Inside network

e. Set ssh timeout for 10 minutes.

16. <u>Configure the Web server</u>

a. Download and install updated VMware on the webserver machine.
b. Create a new virtual machine with Ubuntu install as operating system.
c. Install webmin to the virtual machine.
d. Go to Network > Interfaeces and create new virtual interfaces with eth0 – eth3 and assign ip address , subnet mask, and broadcast address as shown in the table.
e. Test access to the web server

17. <u>Verify the Network</u>

From	Protocol	Destination	Result
PC-1.1	Ping	192.24.104.1	Success
PC-1.1	Ping	PC-2.1	Success
PC-1.1	Ping	PC-3.1	Success
PC-1.1	HTTPS	Web server (Intranet/Internet/B2B)	Success
PC-2.1	HTTPS	Web server (Intranet/Internet/B2B)	Success
PC-3.1	HTTPS	Web server (Intranet/Internet/B2B)	Success
OUTSIDE network from172.24.104.6	HTTP	Internet	Success
OUTSIDE network from172.24.104.x	HTTP	Intranet or B2B	Uusuccess
PC-1.1	SSH	R1	Success
PC-2.1	SSH	ASA INSIDE	Success

18. <u>Troubleshooting</u>

a. **No access to the Router R1**

Found that S1, F0/5 interface assigned to the Vlan 999.

- Configure interface assignment for S1 again with correct interface range.

S1(config)# Interface f0/5
S1(config-if-range)# no shutdown

S1(config)# interface range f0/3 - 4, g0/1 - 2
S1(config-if-range)# switchport mode access
S1(config-if-range)# switchport access vlan 999
S1(config-if-range)# shutdown
S1(config-if-range)# end

- Configure F0/5 trunk interface again.

S1(config)# interface f0/5
S1(config-if)# switchport mode trunk
S1(config-if)# switchport trunk native vlan 1000
S1(config-if)# switchport trunk allowed vlan 20,30,40,1000

b. **Cannot route to the ISP from Inside network**

Found that two default route for R1 configured.

- Remove the 2nd default route by using *no ip route 0.0.0.0 0.0.0.0 172.24.104.1*

c. **Cannot route to the ISP from ASA**.

Found that ASA firewall routing wrongly configured for INSIDE to OUTSIDE

- Remove the existing configuration and added the correct routing.
  NETSEC-ASA(config)# no object network INSIDE-NET

  NETSEC-ASA(config)# object network INSIDE-NET
  NETSEC-ASA(config-object)# subnet 192.168.5.252 255.255.255.252
  NETSEC-ASA(config-object)# nat (INSIDE, OUTSIDE) dynamic interface
  NETSEC-ASA(config-object)# end

d. **DMZ network cannot access any network.**

Found that wrongly configured network mask for S3 virtual interface VALN 50

- Configure correct network mask to the virtual interface.

S3(config)# interface vlan 50
S3(config-if)# ip address 192.168.4.2 255.255.255.192
S3(config-if)# no shutdown

e. **Virtual machines cannot access the INSIDE network.**

Found that wrongly configured default gateway on Virtual machine interface.

- Use *sudo ip route add default via 192.168.4.1 dev ens33* command on terminal to put correct default gateway.
- Then enter *sudo nano /etc/network/interfaces* or *ls /etc/netplan/* command to see the file type.
- Then enter *sudo nano /etc/netplan/01-netcfg.yaml* and correct your default gateway and *Ctrl+o* to Save and *Ctrl+x to* exit.

Or You can do it with webmin

- Go to Webmin and go to Networking > Routing and Gateways > configure correct Default Gateway 192.168.4.1

f. **Virtual machines cannot access the INSIDE network.**
Found that wrong IP configuration for VM interfaces. Then configure ens33 as 192.168.4.4.
– Remove the wrong Ip and configure them all to correct ens33 Ethernet interface.

Device	Interface	IP Address	Subnet Mask	Broadcast address
Virtual Machine	ens33 -Ethernet	192.168.4.4	255.255.255.192	192.168.4.1
	ens33.0 -Intranet	192.168.4.45	255.255.255.192	192.168.4.1
	enss33.1 -Internet	192.168.4.46	255.255.255.192	192.168.4.1
	ens33.2 -B2B	192.168.4.47	255.255.255.192	192.168.4.1

g. **Virtual machines cannot access the INSIDE network.**
Found that wrongly configured DNS client with VM, then configured DNS client as 8.8.8.8
- Use webmin and go to Networking > Hostname and DNS Client > configure correct DNS Server as 8.8.8.8

Budget Estimation

Item	Unit Price ($)	Quantity	Total Price ($)
Router (ISR4321/K9)	1128.00	01	1128.00
Switch (WS-C2960-24TT-L)	625.00	03	1875.00
Firewall (ASA 5506-X)	758.78	01	758.78
Web Server (Beelink Sei8 Mini PC4 Core 8th Gen intel i5-8279U)	473.83	01	473.83
Pcs	749.99	247	185,247.53
Printer (Brother MFC-L3780CDW Color Multifunction Printer)	529.00	03	1587.00
Cables	5.13	700	3591.00
Rack, Casings and Miscellaneous	-	-	5000.00
Total Amount			**199,661.14**

Appendix - 1

Basic Configurations for Network Devices

1. **Router (R1)**
   a. router(config)# hostname R1

   b. R1(config)# no ip domain lookup

   c. R1(config)# enable secret class

   d. R1(config)# line console 0
   R1(config-line)# password cisco
   R1(config-line)# login

   e. R1(config)# line vty 0 4
   R1(config-line)# password cisco
   R1(config-line)# login

   f. R1(config)# service password-encryption

   g. R1(config)# banner motd $ Unauthorized access is Prohibitted! $

   h. R1# copy running-config startup-config

2. **ASA Firewall**
   a. NETSEC-ASA(config)# hostname NETSEC-ASA

   b. NETSEC-ASA(config)# domain-name hsk-spare.com

   c. NETSEC-ASA(config)# password cisco

   d. NETSEC-ASA(config)# banner motd $ Unauthorized access is Prohibitted! $

   NETSEC-ASA(config)# http server enable

   NETSEC-ASA(config)# http 192.168.5.252 255.255.255.252

   e. NETSEC-ASA(config)# write memory

3. **<u>Switches (This is for S1 and configure on all three switches)</u>**
    a. switch(config)# hostname S1
    b. S1(config)# no ip domain lookup
    c. S1(config)# enable secret class
    d. S1(config)# line console 0
       S1(config-line)# password cisco
       S1(config-line)# login
    e. S1(config)# line vty 0 15
       S1(config-line)# password cisco
       S1(config-line)# login
    f. S1(config)# service password-encryption
    g. S1 (config)# banner motd $ Unauthorized access is Prohibitted! $
    h. S1# copy running-config startup-config

4. **<u>Assigning of PCs host address will be done through DHCP server and Assign TFTP server, Business server and Web server addresses as below table.</u>**

Device	**Interface**	**IP Address**	**Subnet Mask**	**Default Gateway**
TFTP Server	**NIC**	**192.168.3.2**	**255.255.255.240**	**192.168.3.1**
Web Server	**NIC**	**192.168.4.3**	**255.255.255.192**	**192.168.4.1**
Business Server	**NIC**	**192.168.2.2**	**255.255.255.192**	**192.168.2.1**

Appendix - 2

List of Each Feature Configured

1. a. **Create VLANs on both switches (This is for S1 and configure on S2 also and for S3 configure Vlan 50, and 999)**

```
S1(config)# vlan 20
S1(config-vlan)# name Management
S1(config-vlan)# vlan 30
S1(config-vlan)# name Operations
S1(config-vlan)# vlan 40
S1(config-vlan)# name Marketing
S1(config-vlan)# vlan 999
S1(config-vlan)# name ParkingLot
S1(config-vlan)# vlan 1000
S1(config-vlan)# name Native
S1(config-vlan)# exit
```

b.

```
S1(config)# interface vlan 30
S1(config-if)# ip address dhcp
S1(config-if)# no shutdown
S1(config-if)# exit
S1(config)# ip default-gateway 192.168.2.1
S1(config)# end

S2(config)# interface vlan 20
S2(config-if)# ip address dhcp
S2(config-if)# no shutdown
S2(config-if)# exit
S2(config)# ip default-gateway 192.168.3.1
S2(config)# end

S3(config)# interface vlan 50
S3(config-if)# ip address 192.168.4.2 255.255.255.192
S3(config-if)# no shutdown
S3(config-if)# exit
S3(config)# ip default-gateway 192.168.4.1
S3(config)# end
```

c.

```
S1(config)# interface range f0/6 - 14
S1(config-if-range)# switchport mode access
S1(config-if-range)# switchport access vlan 30
S1(config-if-range)# exit
S1(config)# interface range f0/15 - 24
S1(config-if-range)# switchport mode access
```

S1(config-if-range)# switchport access vlan 40
S1(config-if-range)# exit

S2(config)# interface range f0/6 - 17
S2(config-if-range)# switchport mode access
S2(config-if-range)# switchport access vlan 20
S2(config-if-range)# exit

S3(config)# interface range f0/1, f0/6
S3(config-if-range)# switchport mode access
S3(config-if-range)# switchport access vlan 50
S3(config-if-range)# exit

d. S1(config)# interface range f0/3 - 4, g0/1 - 2
S1(config-if-range)# switchport mode access
S1(config-if-range)# switchport access vlan 999
S1(config-if-range)# shutdown
S1(config-if-range)# end

S2(config)# interface range f0/3 - 5, f0/18 - 24, g0/1 - 2
S2(config-if-range)# switchport mode access
S2(config-if-range)# switchport access vlan 999
S2(config-if-range)# shutdown
S2(config-if-range)# end

S3(config)# interface range f0/2 - 5, f0/7 - 24, g0/1 - 2
S3(config-if-range)# switchport mode access
S3(config-if-range)# switchport access vlan 999
S3(config-if-range)# shutdown
S3(config-if-range)# end

e. S1# show vlan brief **(This is for S1 and check it on all switches)**

2. <u>Configure Trunking</u>

2.1 Manually configure trunk interfaces F0/1-2 on both Switches (This is for S1 and configure on S2 also)

a. S1(config)# interface range f0/1-2

b. S1(config-if-range)# channel-group 1 mode active
S1(config-if-range)# exit

c. S1(config)# interface port-channel 1

d. S1(config-if)# switchport mode trunk

e. S1(config-if)# switchport mode trunk native vlan 1000

f. S1(config-if)# switchport trunk allowed vlan 20,30,40,1000

g. S1# show interfaces trunk

2.2 Manually configure S1's trunk interface F0/5

a. S1(config)# interface f0/5
S1(config-if)# switchport mode trunk
S1(config-if)# switchport trunk native vlan 1000
S1(config-if)# switchport trunk allowed vlan 20,30,40,1000

b. S1# copy running-config startup-config

c. S1# show interfaces trunk

3. <u>PortFast and BPDU Guard Configuration on Switches (This is for S1 configure on all switches)</u>

a. S1(config)# spanning-tree portpast bpduguard default

4. <u>Configure Inter-Vlan Routing on Router (R1)</u>

a. R1(config)# interface g0/0/1
R1(config-if)# no shutdown

b. R1(config)# interface g0/0/1.20
R1(config-subif)# description Management Network
R1(config-subif)# encapsulation dot1q 20
R1(config-subif)# ip address 192.168.3.1 255.255.255.240
R1(config-subif)# exit
R1(config-subif)# interface g0/0/1.30
R1(config-subif)# encapsulation dot1q 30
R1(config-subif)# description Operations Network
R1(config-subif)# ip address 192.168.2.1 255.255.255.192
R1(config-subif)# exit
R1(config-subif)# interface g0/0/1.40
R1(config-subif)# encapsulation dot1q 40
R1(config-subif)# description Marketing Network
R1(config-subif)# ip address 192.168.1.1 255.255.255.0
R1(config-subif)# exit
R1(config-subif)# interface g0/0/1.1000
R1(config-subif)# encapsulation dot1q 1000 Native
R1(config-subif)# description Native Vlan

b. R1# show ip interface brief

5. **Configure Default Routing**
   a. R1(config)# interface g0/0/0
      R1(config-if)# ip address 192.168.5.253 255.255.255.252
      R1(config-if)# no shutdown
      R1(config-if)# exit

   b. R1(config)# ip route 0.0.0.0 0.0.0.0 g0/0/0 192.168.5.254

6. **Configure DHCP on Router (R1)**
   a. R1(config)#ip dhcp excluded-address 192.168.3.1 192.168.3.2

   b. R1(config)#ip dhcp pool Management
      R1(config-dhcp)# network 192.168.3.0 255.255.255.240

   c. R1(config-dhcp)# dns-server 8.8.8.8

   d. R1(config-dhcp)# default-router 192.168.3.1
      R1(config-dhcp)# domain-name hsk-spare.com

   e. R1#show ip dhcp binding

   Same step to create other configurations for Vlan 30 and Vlan 40

   R1(config)#ip dhcp excluded-address 192.168.2.1 192.168.2.2
   R1(config)#ip dhcp pool Operations
   R1(config-dhcp)# network 192.168.2.0 255.255.255.192
   R1(config-dhcp)# dns-server 8.8.8.8
   R1(config-dhcp)# default-router 192.168.2.1
   R1(config-dhcp)# domain-name hsk-spare.com
   R1(config-dhcp)# end
   R1#show ip dhcp binding

   R1(config)#ip dhcp excluded-address 192.168.1.1
   R1(config)#ip dhcp pool Marketing
   R1(config-dhcp)# network 192.168.1.0 255.255.255.0
   R1(config-dhcp)# dns-server 8.8.8.8
   R1(config-dhcp)# default-router 192.168.1.1
   R1(config-dhcp)# domain-name hsk-spare.com
   R1(config-dhcp)# end
   R1#show ip dhcp binding

7. **Configure R1 for PAT**
   a. R1(config)# access-list 1 permit 192.168.1.0 0.0.0.255
      R1(config)# access-list 1 permit 192.168.2.0 0.0.0.63
      R1(config)# access-list 1 permit 192.168.3.0 0.0.0.15

   b. R1(config)# interface g0/0/0
      R1(config-if)# ip nat outside

   c. R1(config)# interface g0/0/1.20
      R1(config-if)# ip nat inside

      R1(config)# interface g0/0/1.30
      R1(config-if)# ip nat inside

      R1(config)# interface g0/0/1.40
      R1(config-if)# ip nat inside

      R1(config)# interface g0/0/1.1000
      R1(config-if)# ip nat inside

   c. R1(config)# ip nat inside source list 1 interface g0/0/0 overload

8. **Configure all network devices for basic SSH support.**

   8.1 Configure all network devices for basic SSH support. (This is for R1, do this same step on all switches)

   a. R1(config)# username SSHadmin secret cisco12345

   b. R1(config)# ip domain name hsk-spare.com

   c. R1(config)# crypto key generate rsa general-keys modulus 1024

   d. R1(config)# line vty 0 4
      R1(config-line)# transport input ssh
      R1(config-line)# login local
      R1(config-line)# exit

9. **Configure ASA interfaces**
   a. NETSEC-ASA(config)# interface GigabitEthernet1/1
      NETSEC-ASA(config-if)# nameif OUTSIDE
      NETSEC-ASA(config-if)# security-level 0
      NETSEC-ASA(config-if)# ip address 172.24.104.4 255.255.255.248
      NETSEC-ASA(config-if)# no shutdown

   b. NETSEC-ASA(config)# interface GigabitEthernet1/2
      NETSEC-ASA(config-if)# nameif INSIDE

NETSEC-ASA(config-if)# security-level 100
NETSEC-ASA(config-if)# ip address 192.168.5.254 255.255.255.252
NETSEC-ASA(config-if)# no shutdown

c. NETSEC-ASA(config)# interface GigabitEthernet1/3
NETSEC-ASA(config-if)# nameif DMZ
NETSEC-ASA(config-if)# security-level 70
NETSEC-ASA(config-if)# ip address 192.168.4.1 255.255.255.192
NETSEC-ASA(config-if)# no shutdown

10. <u>Configure Default route for the ASA</u>

a. NETSEC-ASA(config)# route OUTSIDE 0.0.0.0 0.0.0.0 172.24.104.1

11. <u>Configure address translation using PAT and network objects</u>

a. NETSEC-ASA(config)# object network INSIDE-NET
NETSEC-ASA(config-object)# subnet 192.168.5.252 255.255.255.252
NETSEC-ASA(config-object)# nat (INSIDE, OUTSIDE) dynamic interface
NETSEC-ASA(config-object)# end

b. NETSEC-ASA(config)# object network DMZ-NET
NETSEC-ASA(config-object)# subnet 192.168.4.0 255.255.255.240
NETSEC-ASA(config-object)# nat (DMZ, OUTSIDE) dynamic interface
NETSEC-ASA(config-object)# end

12. <u>Enable ICMP Inspection in Global Policy</u>

a. NETSEC-ASA(config)# policy-map global_policy
NETSEC-ASA(config)# class inspection_default
NETSEC-ASA(config)# inspect icmp

NETSEC-ASA(config)# show run policy-map

13. <u>Configure Static NAT to the DMZ Servers using a network object</u>

a. NETSEC-ASA(config)# object network INTERNET-SERVER
NETSEC-ASA(config-object)# host 192.168.4.46
NETSEC-ASA(config-object)# nat (DMZ, OUTSIDE) static 172.24.104.6
NETSEC-ASA(config-object)# end

14. **Configure an ACL to allow access to the Web servers from the Internet**
    a. NETSEC-ASA(config)# access-list OUTSIDE-DMZ extended permit ip any host 192.168.4.46
       NETSEC-ASA(config)# access-group OUTSIDE-DMZ in interface OUTSIDE

    b. NETSEC-ASA(config)# access-list DMZ-INSIDE extended permit icmp any any
       NETSEC-ASA(config)# access-list DMZ-INSIDE extended permit tcp any any eq 80
       NETSEC-ASA(config)# access-list DMZ-INSIDE extended permit tcp any any eq 443
       NETSEC-ASA(config)# access-group DMZ-INSIDE in interface DMZ

15. **Configure AAA to use local database for authentication and SSH remote access to ASA**
    a. NETSEC-ASA(config)# username admin password cisco12345

    b. NETSEC-ASA(config)# aaa authentication ssh console LOCAL

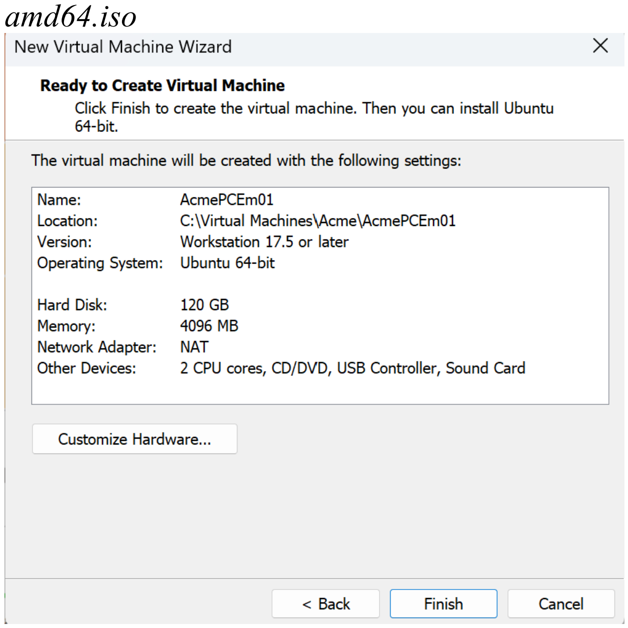
    c. NETSEC-ASA(config)# crypto key generate rsa modulus 2048
       NETSEC-ASA(config)# write mem

    e. NETSEC-ASA(config)# ssh 192.168.5.252 255.255.255.252 INSIDE

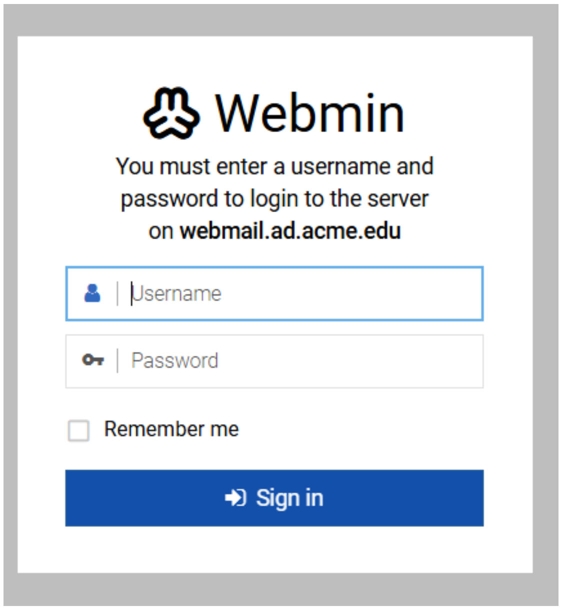
    f. NETSEC-ASA(config)# ssh timeout 10

16. **Configure the Web server**
    a. Download and install *VMware-player-full-17.5.0-22583795.exe*

    b. The details of the virtual machine are as follows, download and install *ubuntu-24.04.2-live-server-amd64.iso*



c. Install webmin.



d. Create new virtual interfaces with eth0 – eth3and details of the interfaces as follows.

Device	Interface	IP Address	Subnet Mask	Broadcast address
Virtual Machine	ens33 -Ethernet	192.168.4.3	255.255.255.192	192.168.4.63
	ens33.0 -Intranet	192.168.4.45	255.255.255.192	192.168.4.63
	ens33.1 -Internet	192.168.4.46	255.255.255.192	192.168.4.63
	ens33.2 -B2B	192.168.4.47	255.255.255.192	192.168.4.63

Appendix – 3
Each Device Configuration

Router (R1)

router(config)# hostname R1
R1(config)# no ip domain lookup
R1(config)# enable secret class
R1(config)# line console 0
R1(config-line)# password cisco
R1(config-line)# login
R1(config)# line vty 0 4
R1(config-line)# password cisco
R1(config-line)# login
R1(config)# service password-encryption
R1(config)# banner motd $ Unauthorized access is Prohibitted! $

R1(config)# interface g0/0/1
R1(config-if)# no shutdown

R1(config)# interface g0/0/1.20
R1(config-subif)# description Management Network
R1(config-subif)# encapsulation dot1q 20
R1(config-subif)# ip address 192.168.3.1 255.255.255.240
R1(config-subif)# exit
R1(config-subif)# interface g0/0/1.30
R1(config-subif)# encapsulation dot1q 30
R1(config-subif)# description Operations Network
R1(config-subif)# ip address 192.168.2.1 255.255.255.192
R1(config-subif)# exit
R1(config-subif)# interface g0/0/1.40
R1(config-subif)# encapsulation dot1q 40
R1(config-subif)# description Marketing Network
R1(config-subif)# ip address 192.168.1.1 255.255.255.0
R1(config-subif)# exit
R1(config-subif)# interface g0/0/1.1000
R1(config-subif)# encapsulation dot1q 1000 Native
R1(config-subif)# description Native Vlan
R1(config-subif)# exit

R1# show ip interface brief

```
R1(config)# interface g0/0/0
R1(config-if)# ip address 192.168.5.253 255.255.255.252
R1(config-if)# no shutdown
R1(config-if)# exit

R1(config)# ip route 0.0.0.0 0.0.0.0 g0/0/0 192.168.5.254

R1(config)#ip dhcp excluded-address 192.168.3.1 192.168.3.2
R1(config)#ip dhcp pool Management
R1(config-dhcp)# network 192.168.3.0 255.255.255.240
R1(config-dhcp)# dns-server 8.8.8.8
R1(config-dhcp)# default-router 192.168.3.1
R1(config-dhcp)# domain-name hsk-spare.com
R1#show ip dhcp binding

R1(config)#ip dhcp excluded-address 192.168.2.1 192.168.2.2
R1(config)#ip dhcp pool Operations
R1(config-dhcp)# network 192.168.2.0 255.255.255.192
R1(config-dhcp)# dns-server 8.8.8.8
R1(config-dhcp)# default-router 192.168.2.1
R1(config-dhcp)# domain-name hsk-spare.com
R1(config-dhcp)# end

R1(config)#ip dhcp excluded-address 192.168.1.1
R1(config)#ip dhcp pool Marketing
R1(config-dhcp)# network 192.168.1.0 255.255.255.0
R1(config-dhcp)# dns-server 8.8.8.8
R1(config-dhcp)# default-router 192.168.1.1
R1(config-dhcp)# domain-name hsk-spare.com
R1(config-dhcp)# end
R1#show ip dhcp binding

R1(config)# access-list 1 permit 192.168.1.0 0.0.0.255
R1(config)# access-list 1 permit 192.168.2.0 0.0.0.63
R1(config)# access-list 1 permit 192.168.3.0 0.0.0.15

R1(config)# interface g0/0/0
R1(config-if)# ip nat outside

R1(config)# interface g0/0/1.20
R1(config-if)# ip nat inside
```

R1(config)# interface g0/0/1.30
R1(config-if)# ip nat inside

R1(config)# interface g0/0/1.40
R1(config-if)# ip nat inside

R1(config)# interface g0/0/1.1000
R1(config-if)# ip nat inside

R1(config)# ip nat inside source list 1 interface g0/0/0 overload

R1(config)# username SSHadmin secret cisco12345
R1(config)# ip domain name hsk-spare.com
R1(config)# crypto key generate rsa general-keys modulus 1024
R1(config)# line vty 0 4
R1(config-line)# transport input ssh
R1(config-line)# login local
R1(config-line)# exit

R1# copy running-config startup-config

<u>ASA Firewall</u>
NETSEC-ASA(config)# hostname NETSEC-ASA
NETSEC-ASA(config)# domain-name hsk-spare.com
NETSEC-ASA(config)# password cisco
NETSEC-ASA(config)# service password-encryption
NETSEC-ASA(config)# banner motd $ Unauthorized access is Prohibitted! $
NETSEC-ASA(config)# http server enable
NETSEC-ASA(config)# http 192.168.5.252 255.255.255.252 INSIDE

NETSEC-ASA(config)# interface GigabitEthernet1/1
NETSEC-ASA(config-if)# nameif OUTSIDE
NETSEC-ASA(config-if)# security-level 0
NETSEC-ASA(config-if)# ip address 172.24.104.4 255.255.255.248
NETSEC-ASA(config-if)# no shutdown
NETSEC-ASA(config)# interface GigabitEthernet1/2
NETSEC-ASA(config-if)# nameif INSIDE
NETSEC-ASA(config-if)# security-level 100
NETSEC-ASA(config-if)# ip address 192.168.5.254 255.255.255.252
NETSEC-ASA(config-if)# no shutdown
NETSEC-ASA(config)# interface GigabitEthernet1/3
NETSEC-ASA(config-if)# nameif DMZ

```
NETSEC-ASA(config-if)# security-level 70
NETSEC-ASA(config-if)# ip address 192.168.4.1 255.255.255.192
NETSEC-ASA(config-if)# no shutdown

NETSEC-ASA(config)# route OUTSIDE 0.0.0.0 0.0.0.0 172.24.104.1

NETSEC-ASA(config)# object network INSIDE-NET
NETSEC-ASA(config-object)# subnet 192.168.5.252 255.255.255.252
NETSEC-ASA(config-object)# nat (INSIDE, OUTSIDE) dynamic interface
NETSEC-ASA(config-object)# end
NETSEC-ASA(config)# object network DMZ-NET
NETSEC-ASA(config)#  subnet 192.168.4.0 255.255.255.192
NETSEC-ASA(config)# nat (DMZ,OUTSIDE) dynamic interface

NETSEC-ASA(config)# policy-map global_policy
NETSEC-ASA(config-pmap)# class inspection_default
NETSEC-ASA(config-pmap-c)# inspect icmp
NETSEC-ASA(config)# show run policy-map

NETSEC-ASA(config)# object network INTERNET-SERVER
NETSEC-ASA(config-object)# host 192.168.4.46
NETSEC-ASA(config-object)# nat (DMZ, OUTSIDE) static 172.24.104.6
NETSEC-ASA(config-object)# exit

NETSEC-ASA(config)# access-list OUTSIDE-DMZ permit ip any host 192.168.4.46
NETSEC-ASA(config)# access-group OUTSIDE-DMZ in interface OUTSIDE

NETSEC-ASA(config)# access-list DMZ-INSIDE extended permit tcp any any eq 80
NETSEC-ASA(config)# access-list DMZ-INSIDE extended permit tcp any any eq 443
NETSEC-ASA(config)# access-list DMZ-INSIDE extended permit icmp any any
NETSEC-ASA(config)# access-group DMZ-INSIDE in interface DMZ
NETSEC-ASA(config)# clear xlate

NETSEC-ASA(config)# username admin password cisco12345
NETSEC-ASA(config)# aaa authentication ssh console LOCAL
NETSEC-ASA(config)# crypto key generate rsa modulus 2048
NETSEC-ASA(config)# write mem
NETSEC-ASA(config)# ssh 192.168.5.252 255.255.255.252 INSIDE
NETSEC-ASA(config)# ssh timeout 10
NETSEC-ASA(config)# write memory
```

Switch (S1)

```
switch(config)# hostname S1
S1(config)# no ip domain lookup
S1(config)# enable secret class
S1(config)# line console 0
S1(config-line)# password cisco
S1(config-line)# login
S1(config)# line vty 0 15
S1(config-line)# password cisco
S1(config-line)# login
S1(config)# service password-encryption
S1 (config)# banner motd $ Unauthorized access is Prohibitted! $
S1 (config)# exit
S1# copy running-config startup-config

S1(config)# vlan 20
S1(config-vlan)# name Management
S1(config-vlan)# vlan 30
S1(config-vlan)# name Operations
S1(config-vlan)# vlan 40
S1(config-vlan)# name Marketing
S1(config-vlan)# vlan 999
S1(config-vlan)# name ParkingLot
S1(config-vlan)# vlan 1000
S1(config-vlan)# name Native
S1(config-vlan)# exit

S1(config)# interface vlan 30
S1(config-if)# ip address dhcp
S1(config-if)# no shutdown
S1(config-if)# exit
S1(config)# ip default-gateway 192.168.2.1

S1(config)# interface range f0/6 - 14
S1(config-if-range)# switchport mode access
S1(config-if-range)# switchport access vlan 30
S1(config-if-range)# exit
S1(config)# interface range f0/15 - 24
S1(config-if-range)# switchport mode access
S1(config-if-range)# switchport access vlan 40
S1(config-if-range)# exit
```

```
S1(config)# interface range f0/3 - 4, g0/1 - 2
S1(config-if-range)# switchport mode access
S1(config-if-range)# switchport access vlan 999
S1(config-if-range)# shutdown
S1(config-if-range)# end

S1# show vlan brief

S1(config)# interface range f0/1-2
S1(config-if-range)# channel-group 1 mode active
S1(config-if-range)# exit
S1(config)# interface port-channel 1
S1(config-if)# switchport mode trunk
S1(config-if)# switchport mode trunk native vlan 1000
S1(config-if)# switchport trunk allowed vlan 20,30,40,1000
S1# show interfaces trunk

S1(config)# interface f0/5
S1(config-if)# switchport mode trunk
S1(config-if)# switchport trunk native vlan 1000
S1(config-if)# switchport trunk allowed vlan 20,30,40,1000

S1# copy running-config startup-config
S1# show interfaces trunk

S1(config)# spanning-tree portpast bpduguard default

S1(config)# username SSHadmin secret cisco12345
S1(config)# ip domain name hsk-spare.com
S1(config)# crypto key generate rsa general-keys modulus 1024
S1(config)# line vty 0 15
S1(config-line)# transport input ssh
S1(config-line)# login local
S1(config-line)# exit
S1(config)# exit

S1# copy running-config startup-config
```

Switch (S2)

switch(config)# hostname S2
S2(config)# no ip domain lookup
S2(config)# enable secret class
S2(config)# line console 0
S2(config-line)# password cisco
S2(config-line)# login
S2(config)# line vty 0 15
S2(config-line)# password cisco
S2(config-line)# login
S2(config)# service password-encryption
S2 (config)# banner motd $ Unauthorized access is Prohibitted! $
S2 (config)# exit
S2# copy running-config startup-config

S2(config)# vlan 20
S2(config-vlan)# name Management
S2(config-vlan)# vlan 30
S2(config-vlan)# name Operations
S2(config-vlan)# vlan 40
S2(config-vlan)# name Marketing
S2(config-vlan)# vlan 999
S2(config-vlan)# name ParkingLot
S2(config-vlan)# vlan 1000
S2(config-vlan)# name Native
S2(config-vlan)# exit

S2(config)# interface vlan 20
S2(config-if)# ip address dhcp
S2(config-if)# no shutdown
S2(config-if)# exit
S2(config)# ip default-gateway 192.168.3.1
S2(config)# end

S2(config)# interface range f0/6 - 17
S2(config-if-range)# switchport mode access
S2(config-if-range)# switchport access vlan 20
S2(config-if-range)# exit

S2(config)# interface range f0/3 - 5, f0/18 - 24, g0/1 - 2
S2(config-if-range)# switchport mode access
S2(config-if-range)# switchport access vlan 999
S2(config-if-range)# shutdown
S2(config-if-range)# end

S2# show vlan brief

S2(config)# interface range f0/1-2
S2(config-if-range)# channel-group 1 mode active
S2(config-if-range)# exit
S2(config)# interface port-channel 1
S2(config-if)# switchport mode trunk
S2(config-if)# switchport mode trunk native vlan 1000
S2(config-if)# switchport trunk allowed vlan 20,30,40,1000
S2# show interfaces trunk

S2(config)# spanning-tree portpast bpduguard default

S2(config)# username SSHadmin secret cisco12345
S2(config)# ip domain name hsk-spare.com
S2(config)# crypto key generate rsa general-keys modulus 1024
S2(config)# line vty 0 4
S2(config-line)# transport input ssh
S2(config-line)# login local
S2(config-line)# exit
S2(config)# exit

S2# copy running-config startup-config

Switch (S3)
switch(config)# hostname S3
S3(config)# no ip domain lookup
S3(config)# enable secret class
S3(config)# line console 0
S3(config-line)# password cisco
S3(config-line)# login
S3(config)# line vty 0 15
S3(config-line)# password cisco

```
S3(config-line)# login
S3(config)# service password-encryption
S3 (config)# banner motd $ Unauthorized access is Prohibitted! $
S3 (config)# exit
S3# copy running-config startup-config

S3(config)# vlan 50
S3(config-vlan)# name DMZ
S3(config-vlan)# vlan 999
S3(config-vlan)# name ParkingLot

S3(config)# interface vlan 50
S3(config-if)# ip address 192.168.4.2 255.255.255.192
S3(config-if)# no shutdown
S3(config-if)# exit
S3(config)# ip default-gateway 192.168.4.1
S3(config)# end

S3(config)# interface range f0/1, f0/6
S3(config-if-range)# switchport mode access
S3(config-if-range)# switchport access vlan 50
S3(config-if-range)# exit

S3(config)# interface range f0/2 - 5, f0/7 - 24, g0/1 - 2
S3(config-if-range)# switchport mode access
S3(config-if-range)# switchport access vlan 999
S3(config-if-range)# shutdown
S3(config-if-range)# end

S3# show vlan brief

S3(config)# spanning-tree portpast bpduguard default

S3(config)# username SSHadmin secret cisco12345
S3(config)# ip domain name hsk-spare.com
S3(config)# crypto key generate rsa general-keys modulus 1024
S3(config)# line vty 0 4
S3(config-line)# transport input ssh
S3(config-line)# login local
S3(config-line)# exit
S3(config)# exit

S3# copy running-config startup-config
```